

Homework 1

- Convert the following base-10 numbers to base-2, base-8 and base-16 manually. Please do it by hand and upload the pictures to the repository.
 1. 123
 2. 254
 3. 503

Homework 2

- Convert the following numbers to base-10 manually. Please do it by hand and upload the pictures to the repository.
 1. 0b1010110
 2. 0xba2e
 3. 0o7532

Homework 3

- Write a Python program that converts the **decimal** number **215** to binary, octal and hexadecimal without using built-in functions or imported modules.

Homework 4

- Calculate the Two's complement of `0b101101111`. Please do it by hand and upload the pictures to the repository.
- Then, write a Python program that does the same.

Homework 5

- Convert the following binary numbers to hexadecimal and octal by hand, using the grouping method (groups of 4 for hexadecimal, groups of 3 for octal). Please do it by hand and upload the pictures to the repository.
- 0b101101110
- 0b11110000

Homework 6

- Given the 8-bit numbers $A = 0b11001010$ and $B = 0b10100110$, compute the following bitwise operations by hand and show your work. Upload the pictures to the repository.
- $A \& B$, $A | B$, $A \wedge B$, $\sim A$ (keep 8 bits)
- For $C = 0b00010110$, compute $C \ll 2$ and $C \gg 1$.

Homework 7

- Write a Python program that, given two integers, prints their AND, OR, XOR, NOT, left shift and right shift results in binary.
- Then: (a) use the & operator with 1 to print whether a number is even or odd, and (b) use a bitmask to clear the lowest 4 bits of 0b10111101.

Homework 8

- Using 8-bit two's complement, give the signed decimal value of the following numbers. Show your work by hand and upload the pictures to the repository.
- 0b10000000
- 0b11111111
- 0b10110100
- Then, state the range of values that a 16-bit signed integer can represent.